Pakhi Banchalia

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Research Interests

AI Safety, Alignment, Reinforcement Learning, Robustness, Multi-objective Optimization, Biologically Inspired Learning

Research Experience

- **Aalto University (AScI Programme)** — *June 2026 - Aug 2026*
  Research Intern (Incoming) — Espoo, Finland
  - Selected for AScI research internship; will study social intrinsic motivation in multi-agent learning and its implications for alignment (with Prof. Guckelsberger)
- **Lossfunk** — *July 2025 - Sept 2025*
  Research Intern — Bangalore, India
  - Built and evaluated homeostatic RL agents in Crafter to study behavior under competing internal drives
  - Analyzed how reward structure variations affect exploration, behavioral stability, and risks of unintended or misaligned policies
  - Compared homeostatic agents to standard RL baselines to evaluate behavioral diversity, robustness, and failure modes under shifting reward structures
- **Computational Machinery of Cognition (CMC) Lab, TU Dresden** — *Nov 2024 - June 2025*
  Research Intern (Independent Collaboration) — Dresden, Germany
  - **Homeostatic Reinforcement Learning:** Designed RL agents with internal physiological drives to study generalization and retention; analyzed how internal regulation affects long-term stability and generalization under changing task distributions.
  - **Dopamine (Reward-Modulated Learning):** Implemented a biologically inspired gradient-free learning algorithm using reward-modulated plasticity; investigated credit assignment without backpropagation via reward-modulated weight perturbations.
- **Google Summer of Code, INCF** — *May 2024 - Sept 2024*
  Contributor — Remote
  - Simulated *C. elegans* neural development using Neural Cellular Automata and evolutionary optimization
  - Modeled evolving neural connectivity using Graph Neural Networks
  - Applied Topological Data Analysis (Kepler Mapper) to analyze developmental and connectivity trajectories
  - Delivered open-source, reproducible code for developmental neural modeling

Publications

- Pillai, H.S., **Banchalia, P.**, Jamboji, V., Krishnan, S.R. (2026). *CAN: Continuously Adapting Network*. In *Data Science and Applications*, ICDSA 2025. Lecture Notes in Networks and Systems, vol 1723. Springer, Cham. [DOI]

Research Projects

- **Homeostatic Crafter**
  - Designed agents governed by multiple internal drives (food, energy, safety, etc.) influencing decision-making
  - Studied how internal regulation affects exploration and behavioral diversity
  - Investigated modular critic architectures for stabilizing multi-objective RL
- **Dopamine: Reward-Modulated Gradient-Free Learning**
  - Contributed to biologically inspired learning rules using reward-modulated weight perturbation
  - Explored alternative credit assignment mechanisms without gradient backpropagation
  - Demonstrated competitive performance in early benchmarks
- **SEANN: Self-Expanding and Adaptive Neural Networks**
  - Designed neural architecture capable of dynamically expanding capacity during continual learning
  - Enabled identification of task-relevant subnetworks to reduce catastrophic forgetting
  - Explored structural plasticity as an alternative to static model scaling

Talks and Posters

- **Brain-like features in homeostatic deep reinforcement learning agents @ Bernstein Conference 2025**
  - Poster presentation on homeostatic RL agents integrating internal and sensory inputs, showing structured behavioral and neural patterns
- **Emergence of Brain-like Features in Homeostatic RL Agents @ BrainBody 2025**
  - Oral presentation; demonstrated coupling between internal states, neural dynamics, and adaptive exploration

Technical Skills

- **Languages:** Python, C++, MATLAB, SQL
- **ML / Research:** PyTorch, TorchVision, Gymnasium (OpenAI Gym), JAX (familiar)
- **Scientific Computing:** NumPy, SciPy, Pandas, Matplotlib, Kepler Mapper (TDA)
- **Research Areas:** Reinforcement Learning, Biologically-Inspired Learning, Neural Cellular Automata, Continual Learning, GNNs
- **Developer Tools:** Git, Linux/Unix, Docker, LaTeX, Weights Biases (WB)

Education

- **Technische Universität Dresden** *Oct 2025 - Present*
  Master of Science in Computational Modeling and Simulation (Applied AI track) Dresden, Germany
  - GPA: not yet available
- **Amrita Vishwa Vidyapeetham Kerala, India** *2021-2025*
  Bachelor of Technology in Computer Science and Artificial Intelligence Amritapuri, India
  - Grade: 8.8/10